

Digital Footprint Analysis & Incident Recovery Simulation

1. Introduction

In today's digital era, organizations leave extensive digital footprints across the internet. These publicly available traces can be leveraged by attackers to plan cyberattacks without direct system access. This project focuses on analysing such digital footprints using Open Source Intelligence (OSINT) techniques and simulating an incident recovery strategy.

2. Objectives

- To collect publicly available information about a target organization
 - To analyse infrastructure, service, and human-related exposure
 - To perform limited and ethical active reconnaissance
 - To assess cybersecurity risks
 - To propose an incident recovery and mitigation plan
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3. Tools Used

- WHOIS Lookup
 - DNS Analysis Tools
 - LinkedIn (OSINT)
 - Nmap
 - Web Browser & Developer Tools
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4. Methodology

4.1 Passive Reconnaissance

Passive reconnaissance was conducted to gather publicly available information without directly interacting with the target systems. This included:

- Domain registration details
- DNS and hosting information

- IP history
- Employee and organizational data from LinkedIn

— Domain Profile

Registrar	MarkMonitor, Inc. MarkMonitor Inc. IANA ID: 292 URL: http://www.markmonitor.com Whois Server: whois.markmonitor.com abusecomplaints@markmonitor.com (p) +1.2086851750	
Registrar Status	clientDeleteProhibited, clientTransferProhibited, clientUpdateProhibited, serverDeleteProhibited, serverTransferProhibited, serverUpdateProhibited	
Dates	8,454 days old Created on 2002-11-02 Expires on 2026-11-02 Updated on 2025-10-01	↗
Name Servers	DNS1.P09.NSONE.NET (has 3,078,570 domains) DNS2.P09.NSONE.NET (has 3,078,570 domains) DNS3.P09.NSONE.NET (has 3,078,570 domains) DNS4.P09.NSONE.NET (has 3,078,570 domains) NS1-42.AZURE-DNS.COM (has 573,724 domains) NS2-42.AZURE-DNS.NET (has 292 domains) NS3-42.AZURE-DNS.ORG (has 161 domains) NS4-42.AZURE-DNS.INFO (has 54 domains)	↗
IP Address	104.18.41.41 - 42 other sites hosted on this server	↗
IP Location	 - Florida - Miami - Cloudflare Inc.	
ASN	 AS13335 CLOUDFLARENET, US (registered Jul 14, 2010)	
Domain Status	Registered And No Website	
IP History	131 changes on 131 unique IP addresses over 21 years	↗
Hosting History	2 changes on 3 unique name servers over 12 years	↗

4.2 Active Reconnaissance

Limited and ethical active reconnaissance was performed to identify exposed services.

Scope Definition:

This phase focused only on identifying publicly accessible services. No exploitation, authentication attempts, or intrusive activities were performed.

5. Findings

5.1 Infrastructure Findings

- Domain registered with MarkMonitor

Registrar Information

Name	Markmonitor Inc.
IANA ID	292
Abuse Contact Email	abusecomplaints@markmonitor.com
Abuse Contact Phone	+1.2086851750

- Long domain age (over 20 years)

Registered On 2002-11-02T15:38:11Z

Expires On 2026-11-02T15:38:11Z

Updated On 2025-10-01T12:53:37Z

- Use of Cloudflare CDN and DNS protection
- Frequent IP changes indicating load balancing
- Strong DNS and hosting configuration

5.2 Service Findings

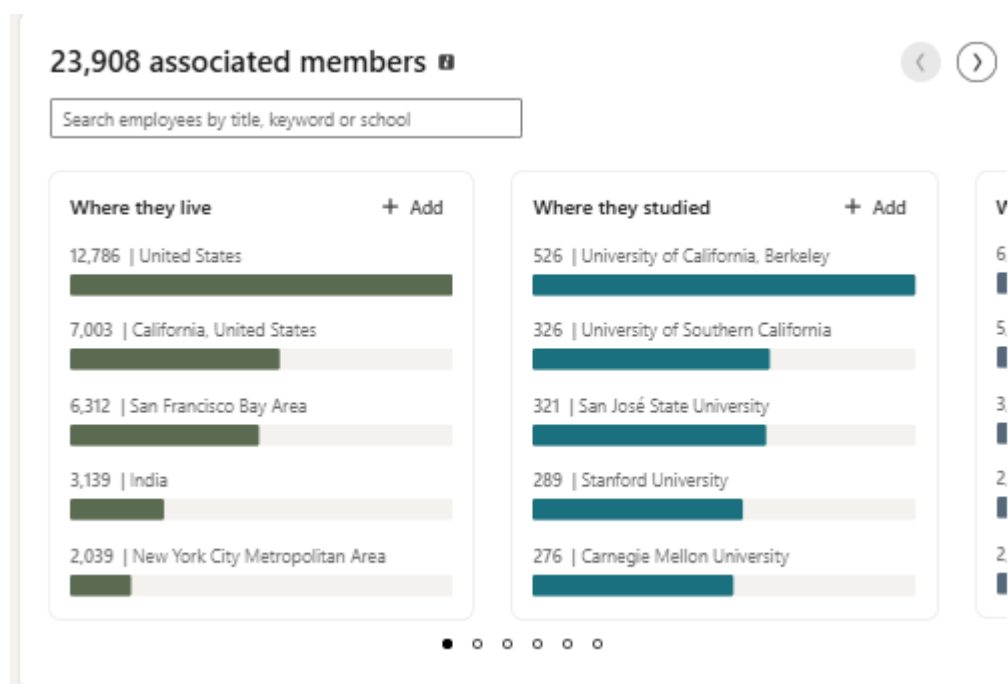
- Open ports:
 - Port 80 (HTTP)
 - Port 443 (HTTPS)
- Majority of ports were filtered
- HTTPS enforced for secure communication

```
C:\Windows\System32>nmap linkedin.com
Starting Nmap 7.98 ( https://nmap.org ) at 2025-12-26 18:24 +0530
Nmap scan report for linkedin.com (150.171.22.12)
Host is up (0.062s latency).
Other addresses for linkedin.com (not scanned): 2620:1ec:50::12
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    closed domain
80/tcp    open  http
443/tcp   open  https

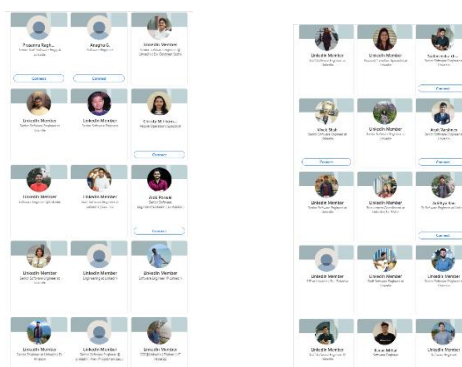
Nmap done: 1 IP address (1 host up) scanned in 9.24 seconds
```

5.3 Human Intelligence Findings

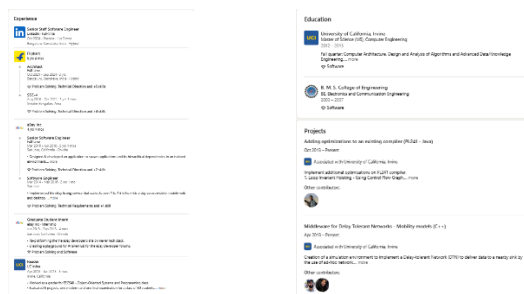
- Large employee base (~23,000+)



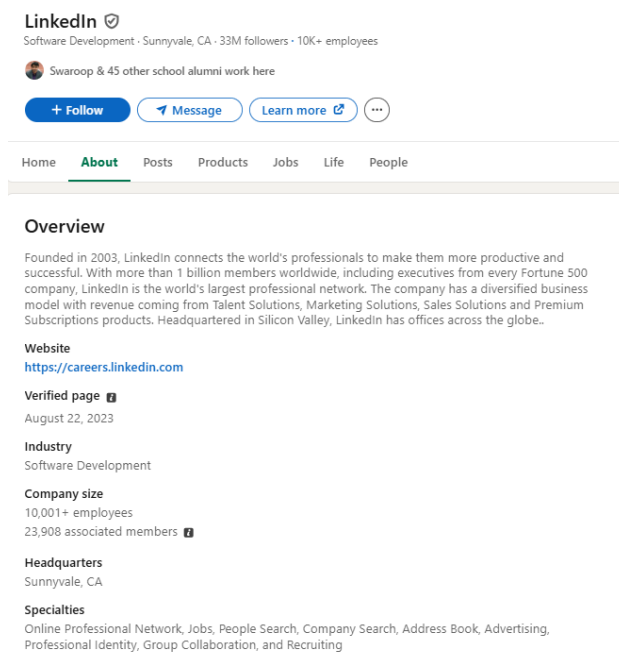
- Publicly available employee roles and locations



- Technology stack mentioned in profiles



- Organizational structure visible



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6. Risk Analysis

6.1 Infrastructure Risk

What you found:

- Cloudflare protection
- Long domain age
- Most ports filtered

How to write it:

The target organization uses enterprise-grade infrastructure protection such as Cloudflare CDN and DNS services. These mechanisms hide the origin servers and reduce direct exposure to attackers.

Risk Level: Low

Reason: Backend systems are not directly accessible from the internet.

6.2 Service Exposure Risk

What you found:

- Only ports 80 (HTTP) and 443 (HTTPS) open
- HTTPS enforced

How to write it:

Active reconnaissance revealed that only essential web services were publicly exposed. No unnecessary or risky services were detected.

Risk Level: Low

Reason: Limited service exposure reduces the attack surface.

6.3 Human & Social Engineering Risk

What you found:

- LinkedIn employee data
- Job roles
- Technologies mentioned

How to write it:

Publicly available employee information such as job roles, locations, and technical skills can be leveraged by attackers to conduct targeted phishing or social engineering attacks.

Risk Level: Medium

Reason: Human factors are often exploited even when technical security is strong.

6.4 Risk Summary Table

Category	Finding	Risk Level	Reason
Infrastructure	Cloudflare protection	Low	Origin masking
Open Ports	Only 80 & 443 open	Low	Minimal attack surface
Network Filtering	Ports filtered	Low	Strong firewall
Human OSINT	Public LinkedIn data	Medium	Social engineering risk

7. Incident Recovery & Mitigation

7.1 Simulated Incident Scenario

An attacker gathers employee information from public sources and launches a phishing attack to gain unauthorized access.

7.2 Incident Response Process

1. **Detection:** Security alerts and phishing reports
2. **Containment:** Account lockdown and blocking malicious sources
3. **Eradication:** Credential resets and removal of malicious content
4. **Recovery:** Restoring access and monitoring systems
5. **Lessons Learned:** Improving detection and awareness

7.3 Preventive Measures

Infrastructure Level

- Continuous port monitoring
- Strong firewall and WAF rules
- Regular SSL audits

Human Level

- Cybersecurity awareness training
 - Limiting excessive public disclosure
 - Phishing simulation exercises
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8. Conclusion

This project demonstrated how OSINT techniques can be used to analyse an organization's digital footprint. While the target organization showed strong infrastructure security, human-related information exposure presents potential risks. Proper awareness, monitoring, and preventive strategies are essential for maintaining a robust security posture.

9. References

- WHOIS Databases – Accessed December 2025
 - LinkedIn Public Profiles – Accessed December 2025
 - Nmap Documentation – <https://nmap.org>
 - OSINT Framework - <https://osintframework.com>
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Note: *“This project was conducted strictly for academic purposes using publicly available information and non-intrusive reconnaissance techniques.”*

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Project Type:

OSINT-Based Cybersecurity Analysis